



EEG-Based Depression Detection Using Multiple Machine Learning Techniques

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Abstract

Depression diagnosis via EEG is challenging due to signal complexity and non-stationarity. This study detects depression patterns from EEG using **multiband analysis**, **CNN**, **XGBoost**, and **Random Forest**^[1].

- Dataset: MODMA (128-channel resting-state EEG)^[1]
- Classes: MDD vs. Healthy Control (6 MDD subtypes)
- Best: **CNN achieved 97% accuracy** at 25 epochs^[1]
- Demographic factors (age, gender) integrated with EEG features

MDD Classification Categories

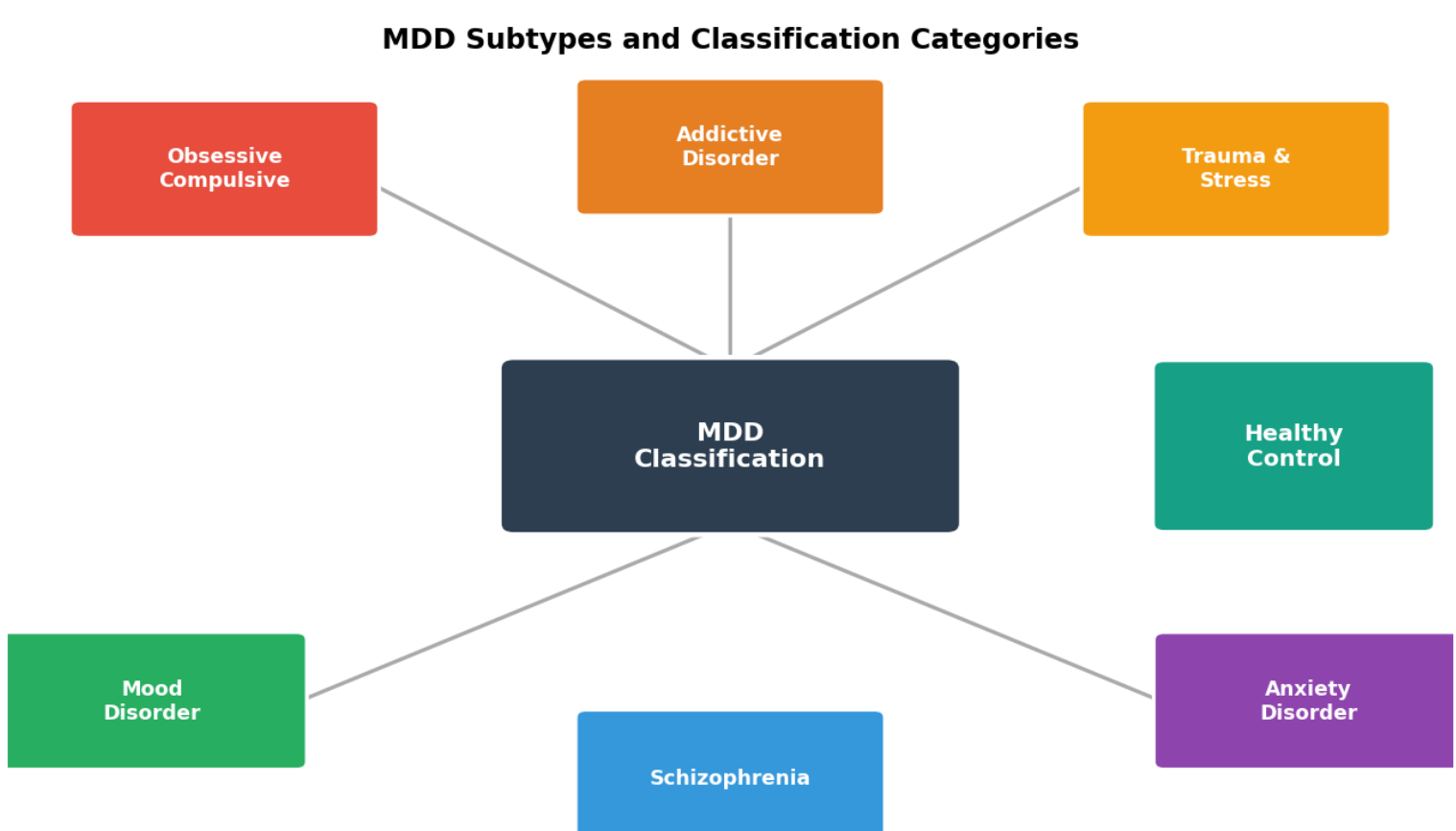


Figure 1. MDD subtypes and Healthy Control

Patient status divided into **MDD** and **Healthy Control**^[1]. MDD includes six subtypes: Obsessive-Compulsive, Addictive Disorder, Trauma & Stress, Mood Disorder, Schizophrenia, Anxiety Disorder.

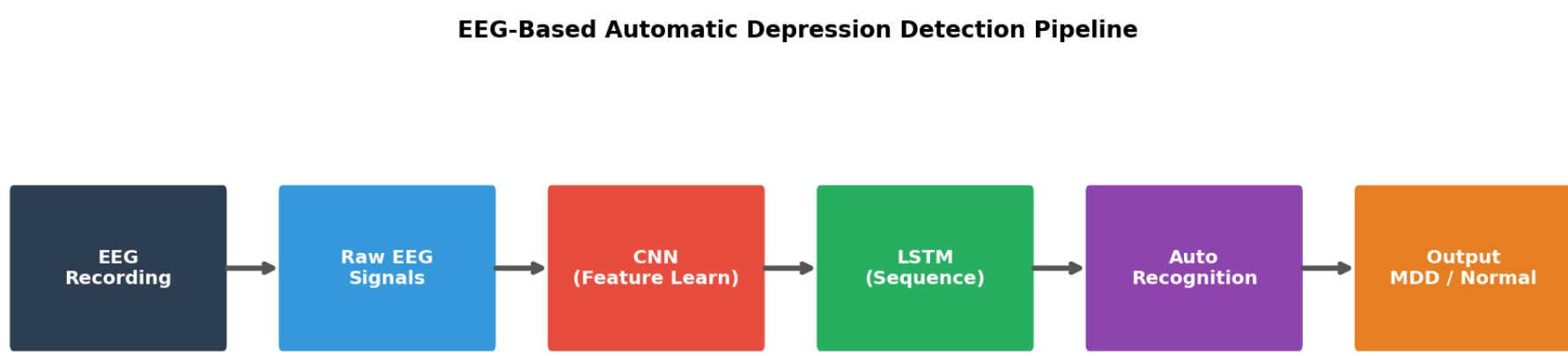
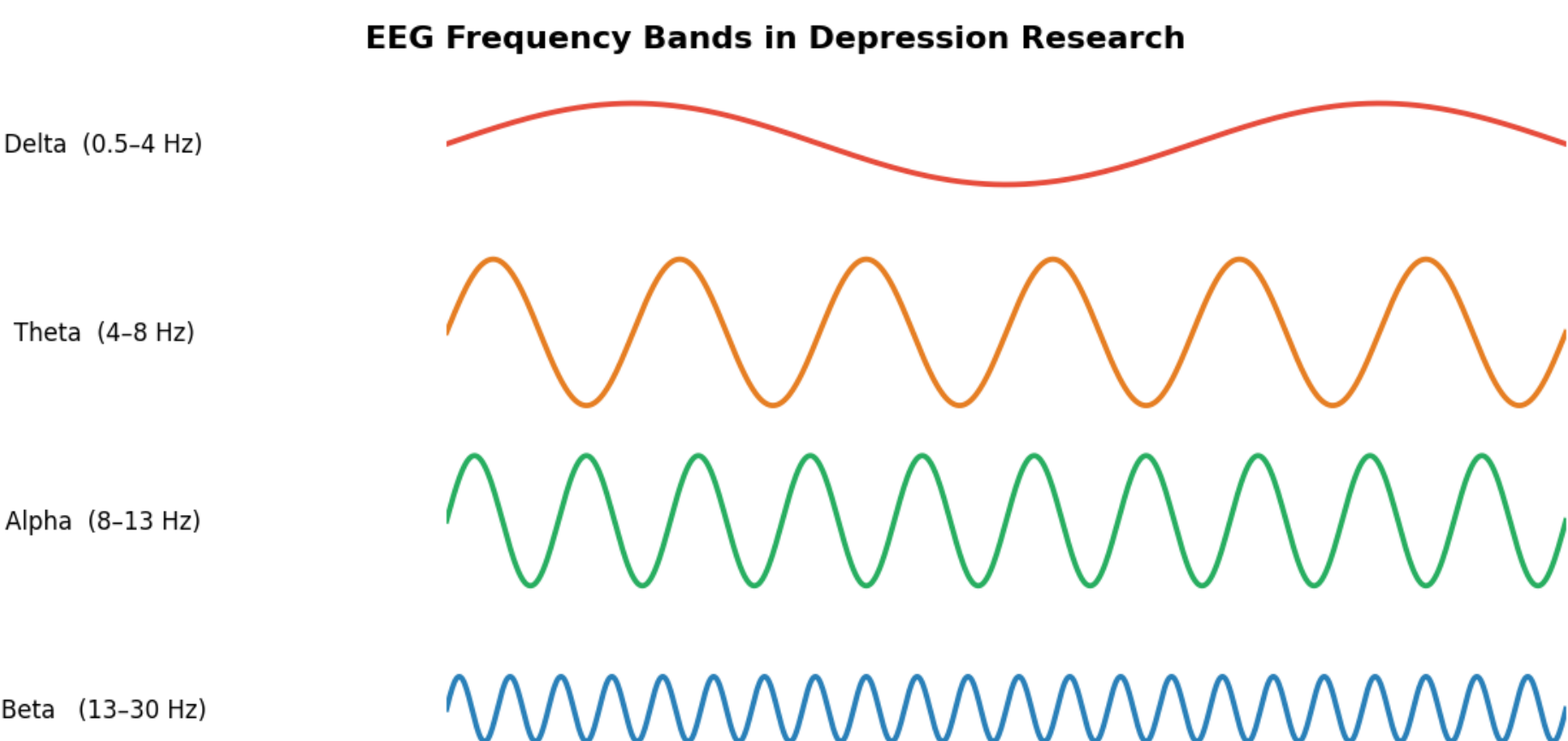


Figure 2. EEG-based depression detection pipeline

EEG Frequency Bands



Dataset: MODMA

MODMA provides EEG recordings via a 128-electrode elastic cap and a wearable 3-electrode collector^[1]. This project uses **128-channel resting-state recordings** (24 MDD + 29 controls).

MODMA Dataset Description				
Experiment	Participants	MDD (M/F)	Controls (M/F)	Age
ERP 128-ch	53	13M/11F	20M/9F	16-56 yrs
3-ch Resting	55	15M/11F	19M/10F	16-56 yrs
128-ch Resting	53	16M/8F	20M/9F	16-56 yrs

Figure 4. MODMA dataset composition

- Inclusion:** Age 18–55; PHQ-9 ≥ 5 ; no psychotropic drugs in last 2 weeks; written consent^[1]
- Exclusion:** Brain damage; mental disorder history; alcohol/drug dependence; pregnancy^[1]
- ANOVA:** No significant age difference ($p > 0.05$) – changes reflect depression, not age^[1]
- Recording:** Eyes closed during EEG capture to reduce visual artefacts; constant lighting maintained^[1]

Pre-processing Pipeline

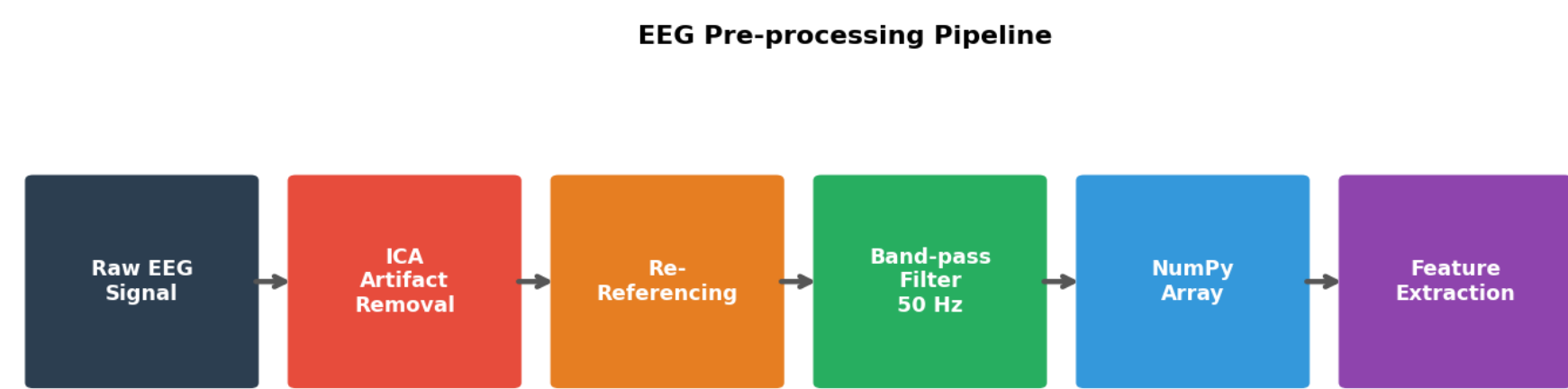


Figure 5. EEG pre-processing workflow

Raw EEG signals must be cleaned before feature extraction^[1]:

- ICA Artifact Removal** – separates brain sources from muscle, eye-blink and cardiac artefacts^[3]
- Re-Referencing** – minimises noise unrelated to brain activity; improves signal-to-noise ratio (SNR)^[1]
- Band-pass Filter (50 Hz)** – removes AC power-line spikes; signal converted to NumPy array^[1]
- Feature Extraction** – linear and non-linear features (spectral entropy, SVD entropy) merged with demographic data^[1]
- Signal Segmentation** – continuous EEG divided into fixed-length epochs to enable consistent windowed feature extraction across all channels^[3]

Classification Models

Three models trained per MDD subtype, 10-fold cross-validation^[1]:

(1) **Random Forest** – Decision Tree ensemble on bootstrap samples; majority vote output. Estimators: [100, 300, 500]^[1].

(2) **XGBoost** – parallel gradient-boosted trees with regularisation. Sub-sample: [0.3, 0.5, 1.0]^[1].

(3) **1D-CNN** – Conv1D (kernel=11, ReLU) → Dropout(0.2) → Max-Pool(4) → Dense(100) → Sigmoid^[1]. Epochs=25, Batch=32, LR=0.001.

Evaluation Metrics: Accuracy, Precision, Recall, F1 Score (Micro & Macro), and ROC-AUC^[4].

Cross-Validation: 10-fold stratified validation ensures results generalise across unseen patient data and reduces bias from dataset imbalance^[4].

CNN Architecture & Training

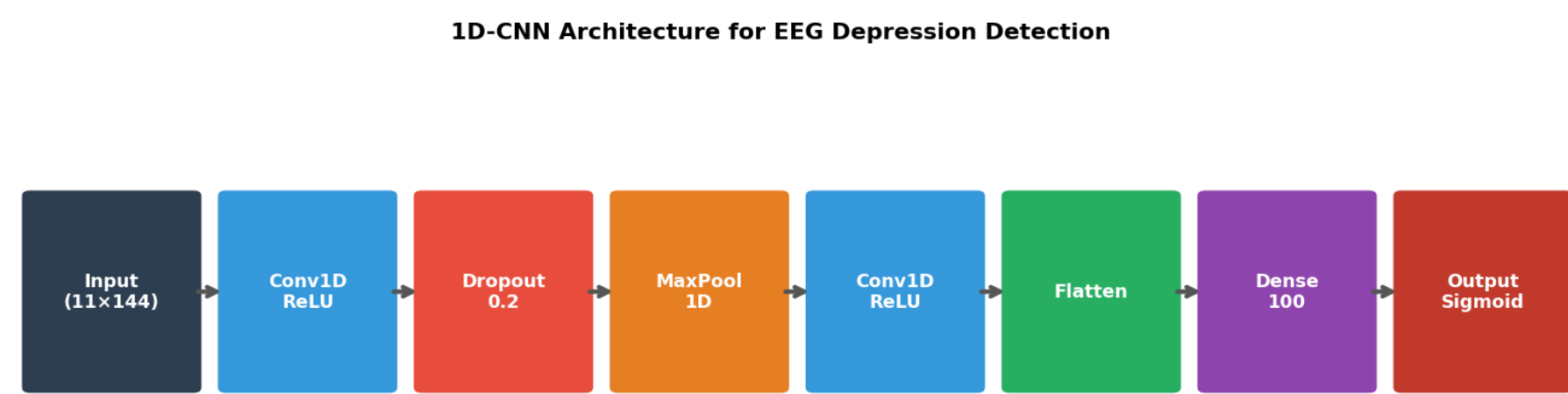


Figure 6. 1D-CNN layer architecture

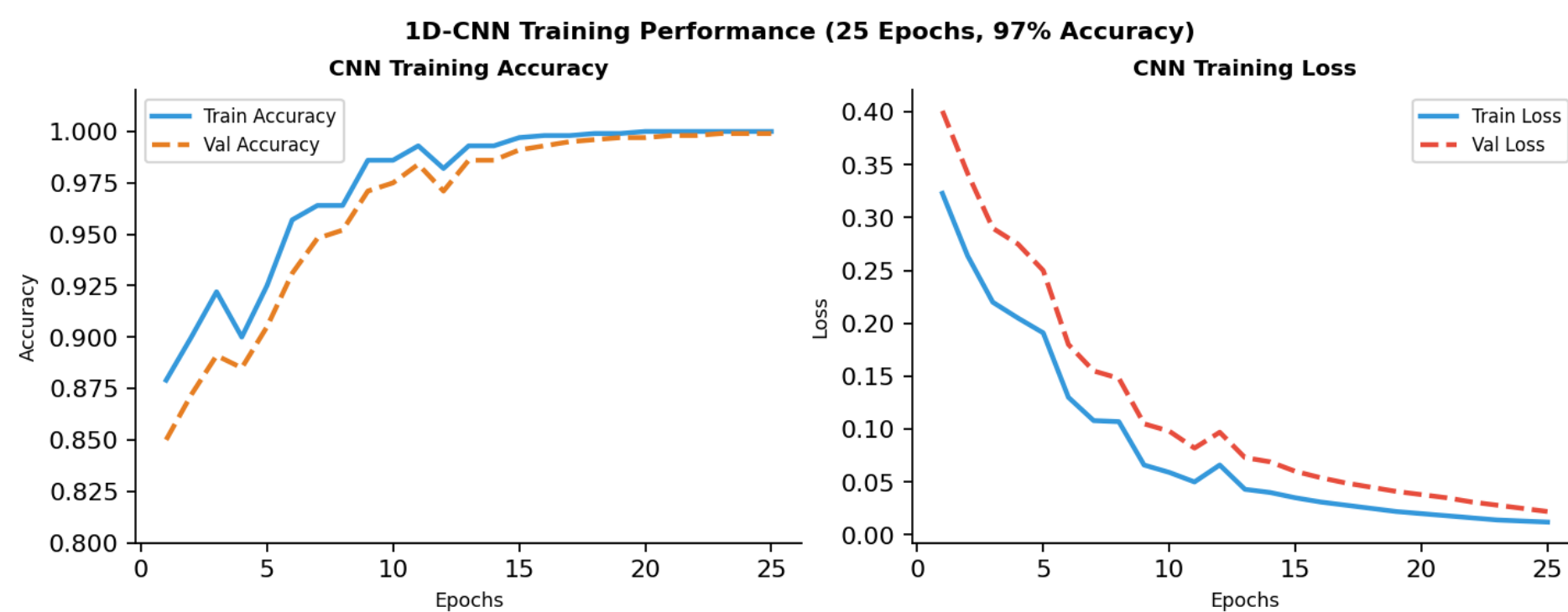


Figure 7. Accuracy and loss over 25 epochs

- Accuracy: 87.9% (epoch 1) → **99.3%** (epoch 25)^[1]
- Loss: 0.323 → **0.043** over 25 epochs^[1]
- Train and validation curves converge – no overfitting^[1]
- CNN outperforms RF and XGBoost on all six MDD classes^[1]
- Binary Cross-Entropy loss** as objective function; sigmoid gives probability output^[2]
- Dropout (0.2)** prevents overfitting by deactivating 20% of neurons per step^[2]
- MaxPooling (size=4)** reduces dimensionality and retains dominant temporal features^[2]
- Dense layer (100 units, ReLU)** learns high-level combinations of EEG features^[2]

Results: Accuracy Across MDD Classes

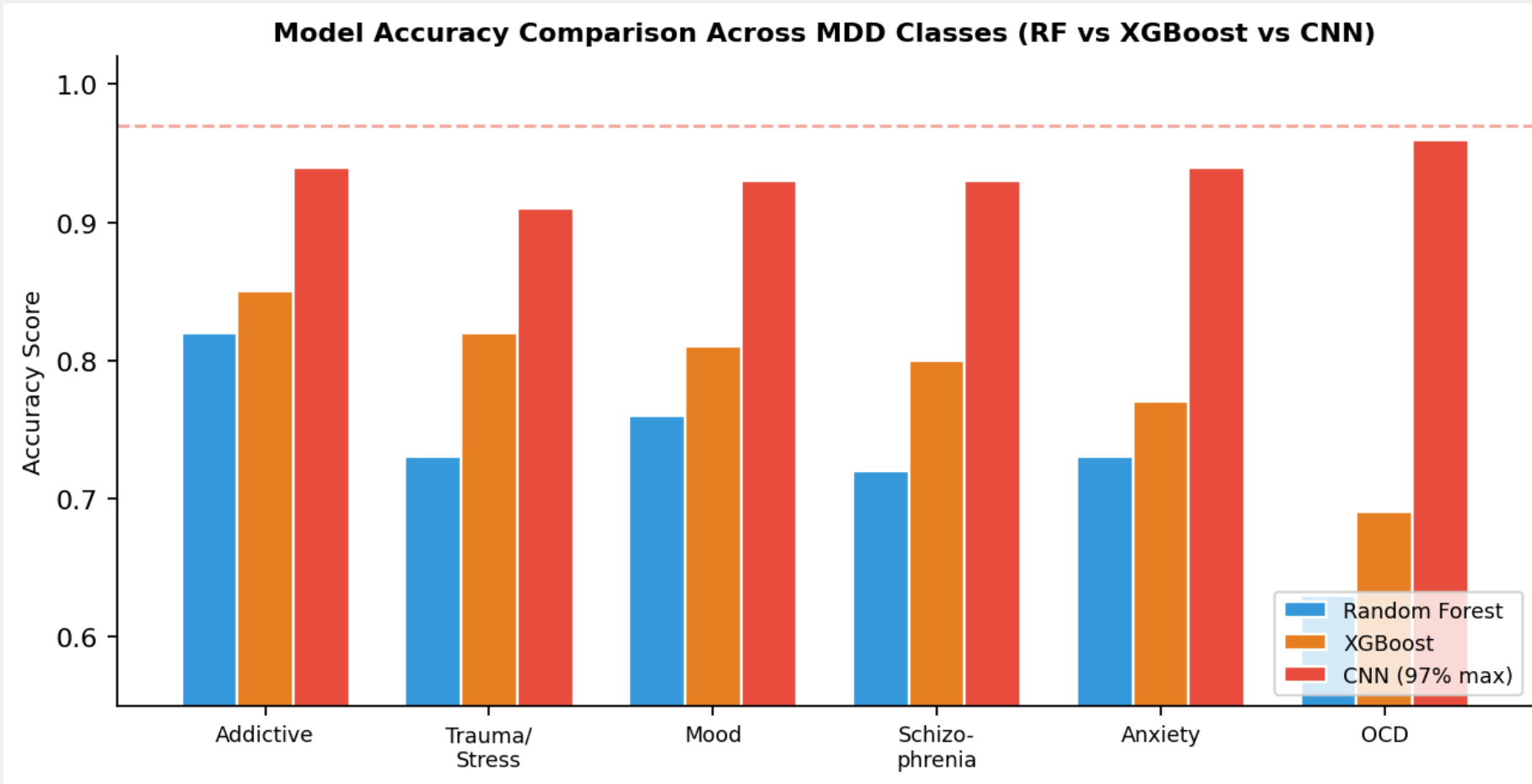


Figure 8. RF vs. XGBoost vs. CNN per MDD subtype

Disorder	RF	XGB	CNN	Best
Addictive	0.82	0.85	0.94	CNN
Trauma/Stress	0.73	0.82	0.91	CNN
Mood	0.76	0.81	0.93	CNN
Schizophrenia	0.72	0.80	0.93	CNN
Anxiety	0.73	0.77	0.94	CNN
OCD	0.63	0.69	0.96	CNN

Conclusion

- CNN outperformed** RF and XGBoost across all 6 MDD classes with up to **97% accuracy** at 25 epochs^[1]
- Combining **EEG signals with demographic data** (age, gender) improves detection precision^[1]
- MODMA** with 128-ch and 3-ch collectors demonstrates real-world applicability of the system^[1]
- Future work:** collect more data, implement severity grading (mild/moderate/severe), adopt lightweight ear-BCI electrodes for practical deployment^[3]
- Future work:** explore explainable AI (XAI) techniques to improve clinical trust and model interpretability^[4]
- Future work:** extend to real-time mental state monitoring using edge computing and wearable devices^[6]

References

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